

Question	Answer	Mark	Mark scheme	Additional guidance
20	$1 + \frac{\sqrt{5}}{5}$	P1 P1 P1 A1	for writing $\sqrt{180}$ as $6\sqrt{5}$ for process to rationalising the denominator eg $\frac{\sqrt{180} - 2\sqrt{5}}{5\sqrt{5} - 5} \times \frac{5\sqrt{5} + 5}{5\sqrt{5} + 5}$ or $\frac{4\sqrt{5}}{5\sqrt{5} - 5} \times \frac{5\sqrt{5} + 5}{5\sqrt{5} + 5}$ oe (dep on previous P1) for expanding terms eg $\frac{5\sqrt{5}\sqrt{180} + 5\sqrt{180} - 50 - 10\sqrt{5}}{125 - 25}$ or $\frac{100 + 20\sqrt{5}}{100}$ oe for $1 + \frac{\sqrt{5}}{5}$	This process mark can be awarded whenever this is seen, which might be later in the process. Accept written as $a = 1, b = 5$